Gamblers Ruin

Gareth Tribello[1]

[1]School of Mathematics and Physics

Keywords: Gamblers ruin, Random walk, Dependent and independent variables

14th August, 2024

Abstract

A simple model that can be used to simulate how gamblers play games like roulette has been introduced. The model assumes that a gambler enters casinos intending to win a certain amount of money and that they will keep playing until they either win the amount of money they desire or until they lose all their money. We use our model to investigate how the number of games the gambler plays depends on the amount of money they have when they enter the casino.

1 Introduction

In this report, I am going to study the problem known as Gamblers ruin using computer simulation. The details of this problem are as follows:

- A gambler has k pounds with which to play.
- He would like to win $N > k$ pounds.
- He places a stake of 1 pound on an event with a probability p of occurring.
- If the event occurs, he wins back 2 pounds and thus has $k + 1$ pounds.
- If the event does not occur, he loses his stake and thus has $k - 1$ pounds.
- He repeats this process of betting 1 pound on this event until he is left with 0 pounds or until he wins the N pounds he desires.

It is straightforward to design a mathematical model that describes how the amount of money the gambler has depends on time. This model should include some element of randomness because the outcome of each game is unknown. We should, nevertheless, be able to use this model to predict how likely the gambler is to leave the casino with 0 pounds or how many games he would expect to play.

This report aims to determine how the expected number of games the gambler will play depends on the amount of money he has when he enters the casino. With this in mind, the report is thus structured as follows. In section 2 we discuss the function that we have used to simulate how much money the gambler has when he leaves the casino and how many times he will have played. In section 3 this function is then used to determine the probability of leaving the casino with N pounds. We then investigate the distribution for the number of turns the gambler will have in the casino in section 4. The report then finishes in section 5 by determining how the expected number of games the gambler will play depends on the amount of money he has when he enters the casino.

2 Simulating gamblers ruin

The cell below shows the function that we have used to simulate the gamblers ruin problem. This function takes N, k and p as its input parameters. It returns two numbers:

- The first of these numbers `outcome` is equal to 1 if he finished with N pounds and is equal to zero if he finished with 0 pounds
- The second of these numbers `nturns` is equal to the number of times that he played the game.

```python
import numpy as np

def gambler( N, k, p ) :
    # Check initial position is valid
    assert( k>0 and k<N )
    nturns = 0   # This variable will count the number of turns
    # Generate new turns until the gambler arrives in state 0 or state N.
    while( k>0 and k<N ) :
        # Note that one additional turn has been taken
        nturns = nturns + 1
        # Move forward with probability p
        if np.random.uniform(0,1)<p : k = k + 1
        else : k = k - 1

    outcome = 1
    if k==0 : outcome=0
    else : assert( k==N )
    return outcome, nturns
```

Listing 1: Simulating gamblers ruin

3 Probability of winning

I used the function that was introduced in section 2 to calculate the probability that a gambler who enters the casino with 5 pounds to his name and a desire to have ten pounds wins the ten pounds. I supposed that he has a probability of 0.5 of winning each of the individual games that he plays. To estimate the probability of his winning the ten pounds, I modelled the process of him playing the 500 times and calculated the mean and variance from these 500 samples of `outcome`. I am thus 90% certain that the probability of him winning the 10 pounds is between 0.425 and 0.499.

4 Distribution for length of game

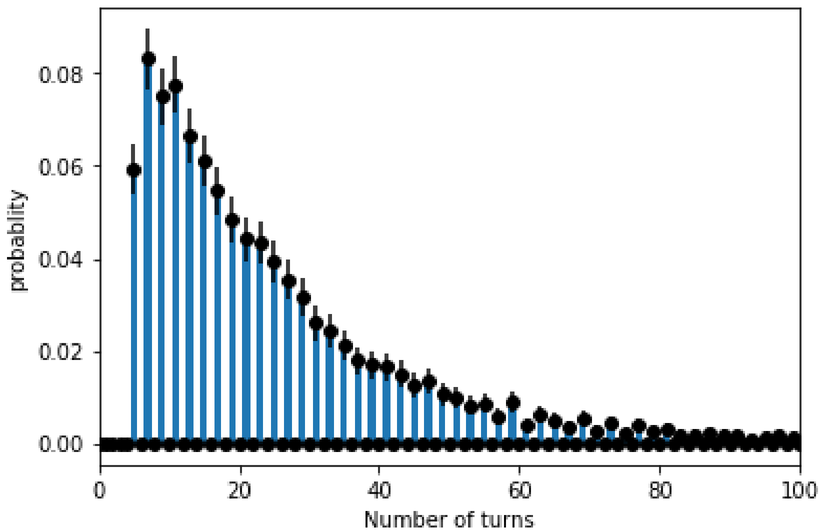

Fig. 1. An estimate of the probability mass function for the number of turns the gambler takes. The centres of the error bars and the heights of the bars indicate the fraction of times each result came up in a set of 5000 samples. The error bars indicate a 90% confidence limit on the estimate of the probability mass function.

The number of turns the gambler takes before he loses all his money or wins his desired target is a discrete random variable. Figure 1 shows an estimate of the probability mass function for this random variable when $p = 0.5$, $N = 10$ and $k = 5$. This estimate was generated by taking 5000 samples from the distribution. The error bars in this figure illustrate the 90% confidence limit on the estimate of the probability mass function.

5 Dependence of length of game on start point

Figure 2 shows how the average length of the game depends on the amount of money that the gambler starts with, k. Each of the 9 averages that are shown in figure 2 was calculated from 2000 samples of the random variable with $N = 10$ and $p = 0.5$. The error bars indicate the 90% confidence limit on the estimates of the averages that have been obtained by sampling.

Figure 2 shows that the number of turns the gambler plays is maximised when he starts with 5 pounds, i.e. at the midpoint between the two endpoints. Perhaps unsurprisingly, he plays the smallest number of games when he starts with 1 or 9 pounds. In this case, because the probability of moving forward one step is equal to the probability of moving backwards,

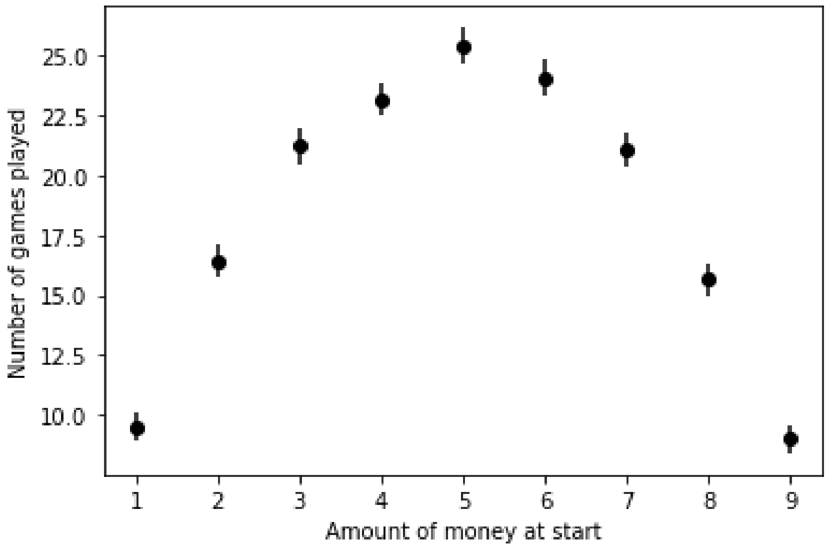

Fig. 2. The dependence of the number of games played on the amount of money the gambler starts with. Each of the averages that are plotted on the y-axis of this graph was calculated by taking 2000 samples. Error bars indicate the 90% confidence limit on these averages.

the graph above is symmetric around $k = 5$. Any loss of this symmetry can be attributed to random error.